STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 2 -

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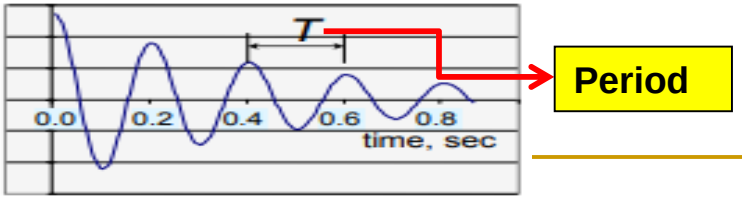
LOADS

In general, the effects causing internal forces and displacements in structures are referred to as loads.

There are two main types of loads as *static* and *dynamic*.

The most significant parameter, which defines a load as static or dynamic, is the natural period of vibration of the structure.

Note: Natural period of vibration is the time required for the structure to go through one complete cycle of free vibration.



Static Load: If the application time for the load is large compared to the natural period of the structure, then there will be no dynamic effect and the load can be considered static.

Dynamic Load:If, on the other hand, the application time for the load is in close proximity to the natural period of the structure, it will induce a dynamic response.

The term *dynamic* may be defined simply as *time-varying*.

Common types of loads considered in designe of a structure:

- Dead load *(Static load)*
- Live load *(Can be considered as static)*
- Earthquake load *(Dynamic load)*
- Wind load *(Dynamic load)*
- Snow load *(Static load)*
- Hydrostatic and Soil Pressure *(Static load)*
- Temperature change, shrinkage, creep *(Static load)*

Dead Load

Dead loads consist of the weights of the various structural members and the weights of any objects that are permanently attached to the structure. It is denoted by «g» or «G».

As an example, dead load of a building consists of,

- columns, beams, floor slab

- roofing

- infill walls, windows

- plumbing

- electrical fixtures

- other miscellaneous attachments

Live Load

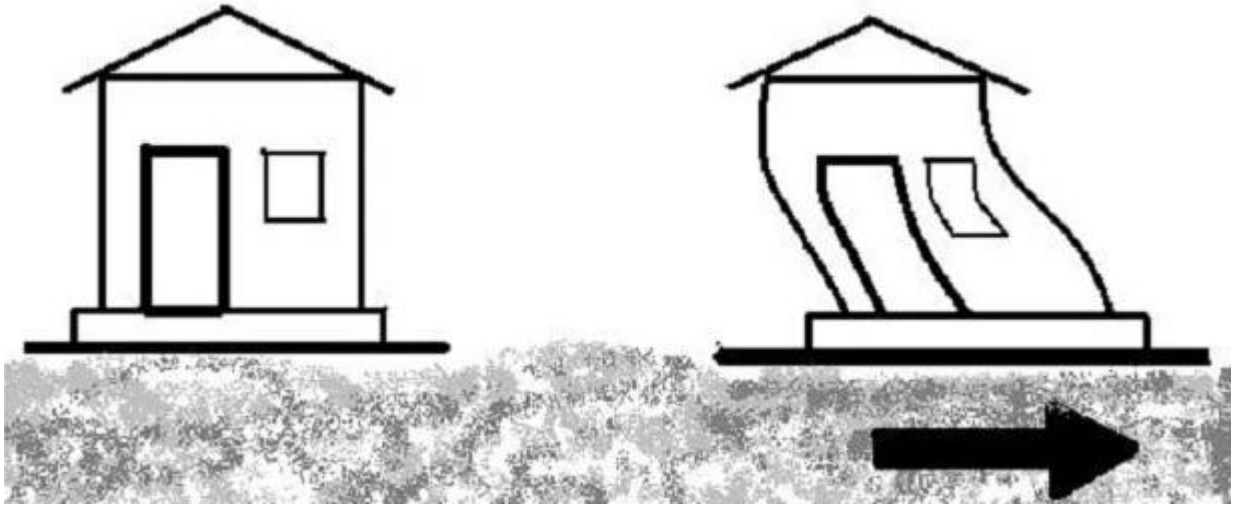
Live loads can vary both in their magnitude and location. They may be caused by the weights of objects temporarily placed on a structure, or moving vehicles. The minimum live loads specified in codes are determined from studying the history of their effects on existing structures. Live load is denoted by «q» or «Q».

For example,

- *Q = 2* kN/m^2 (for *house rooms)*
- *Q = 3.5* kN/m^2 (for house stairs)
- *Q = 5* kN/m^2 (for library)

Earthquake Load

Earthquakes produce loadings on a structure through its interaction with the ground and its response characteristics.

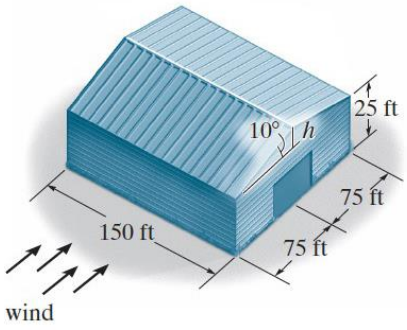


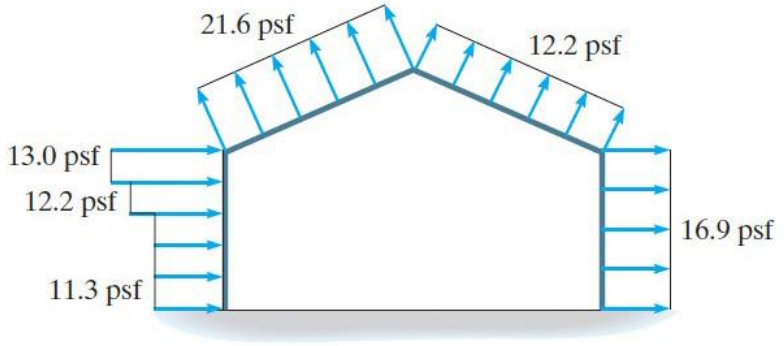
Wind Loads

When structures block the flow of wind, the wind's kinetic energy is converted into potential energy of pressure, which causes a wind loading.

Although wind load is a dynamic load, it can be considered as static load.







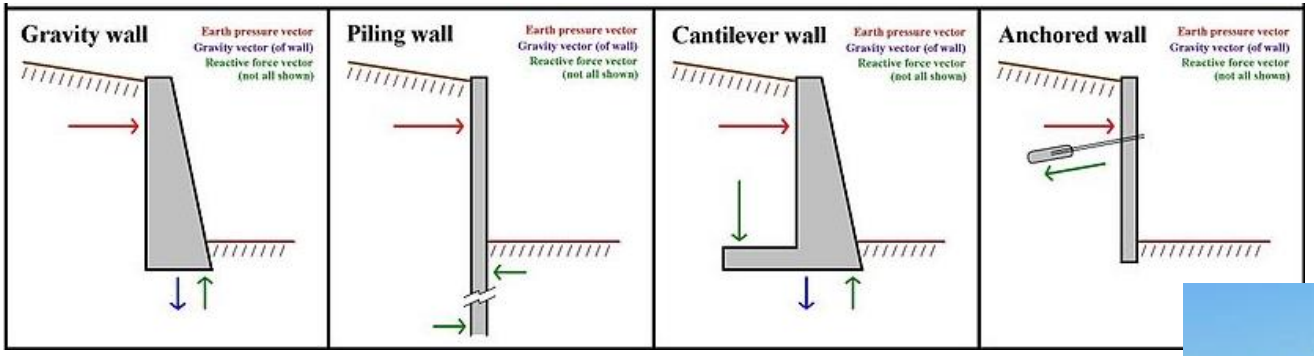
Snow Loads

In some parts of the country, roof loading due to snow can be quite severe, and therefore protection against possible failure is of primary concern.



Hydrostatic and Soil Pressure

When structures are used to retain water, soil, or granular materials, the pressure developed by these loadings becomes an important criterion for their design. Examples of such types of structures include tanks, dams, ships, bulkheads, and retaining walls.



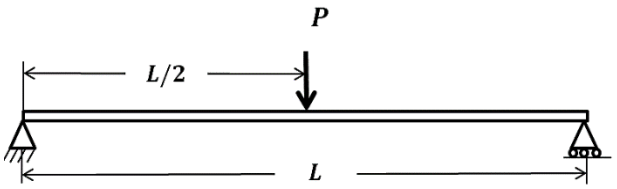


Temperature, Shrinkage, Creep

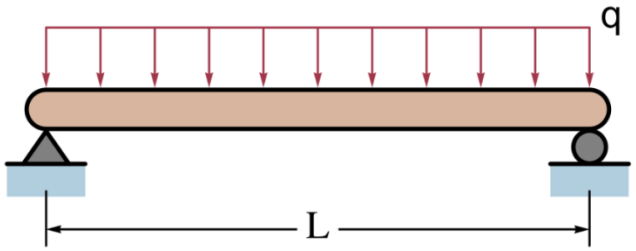
Those effects cause changes in the dimensions of structural elements. If these changes are restrained by support conditions, then there occur internal forces in the structural elements.

Idealization of Loads

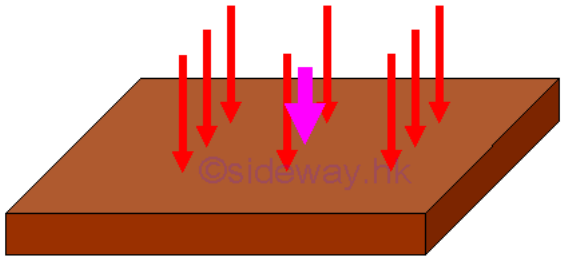
- Point loads



- Distributed loads



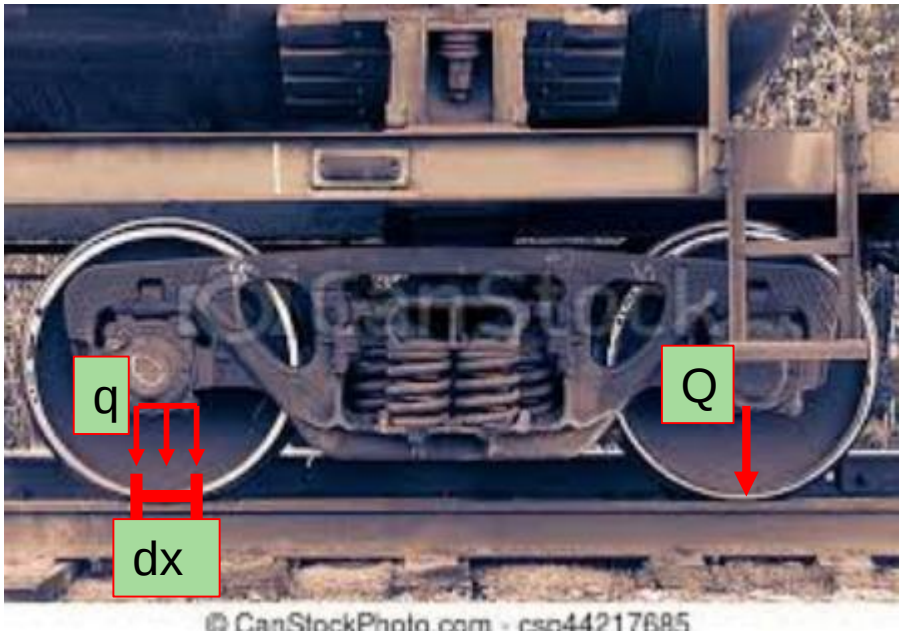
Line Load

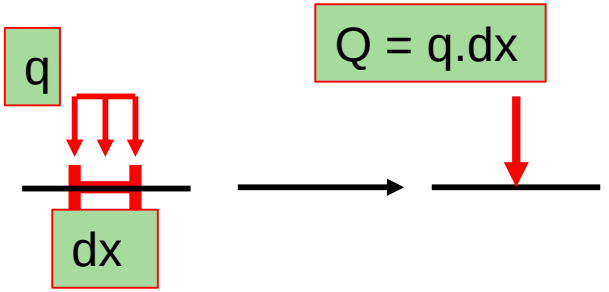


Area Load

Point loads

A load acting at a point of structure is referred to as point load. In reality, there does not exist point loads, however a load acting on a small area or a short distance is considered as a point load in the analysis. It is the resultant of a load acting on the small area or short distance.

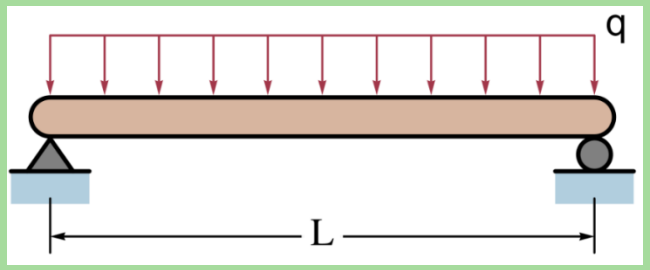




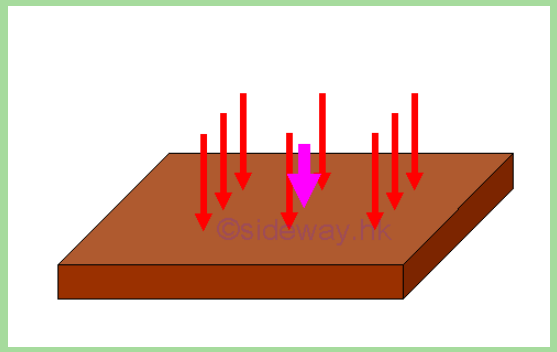
Distributed loads

A load acting on a certain length or area is referred to as a distributed load.

If it acts over a length, it is referred to as a line load.

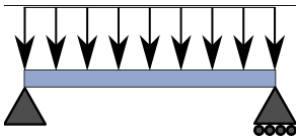


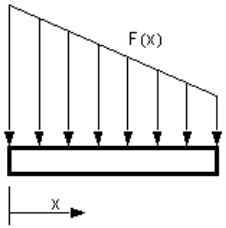
If it acts on an area, it is referred to as an area load.

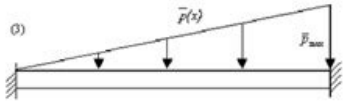


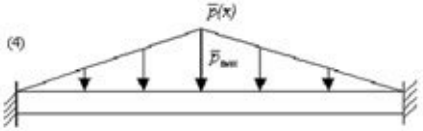
Distributed loads

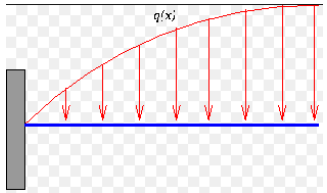
- Regularly distributed loads
  - Uniformly
  - Trapezoidal
  - Triangular
  - Parabolic
- Irregularly distributed loads

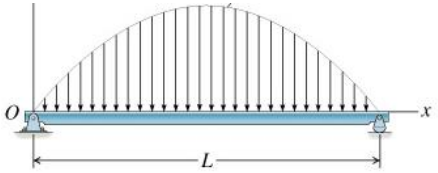


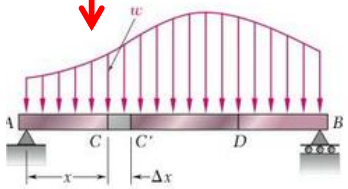






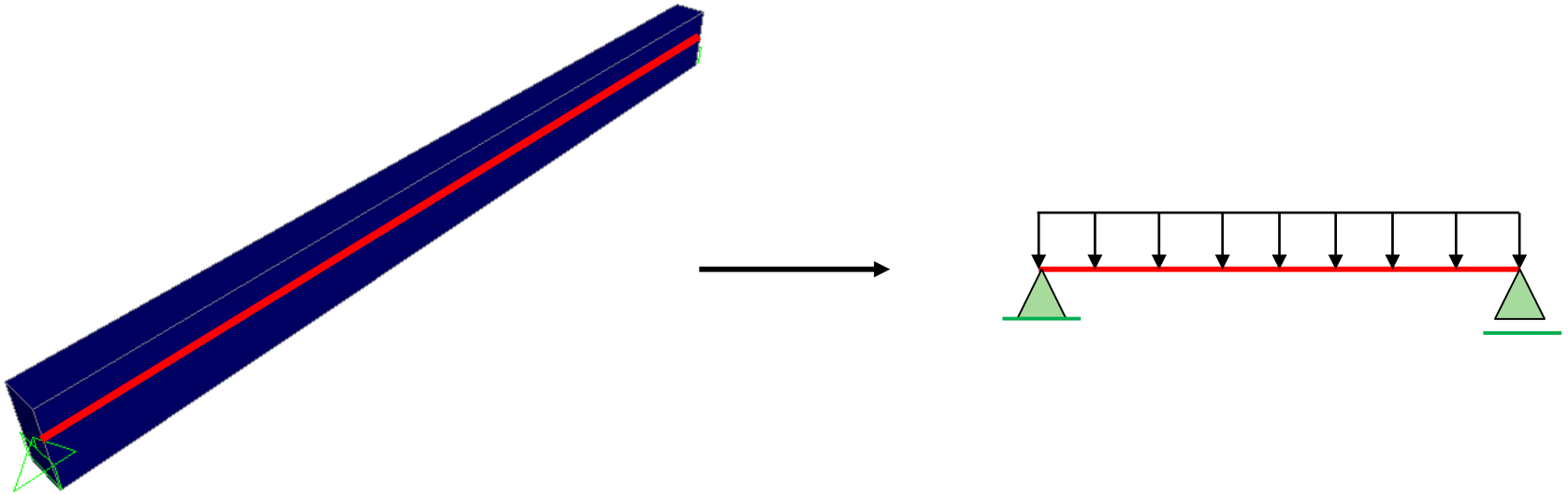






Uniformly distributed load

A distributed load with a constant magnitude is uniformly distributed load.

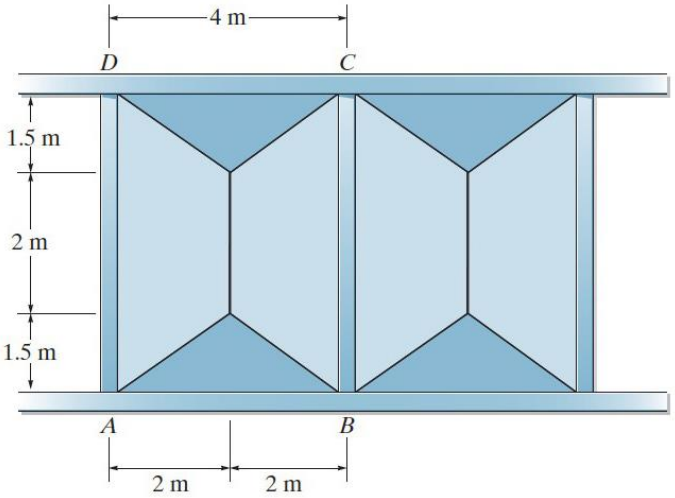


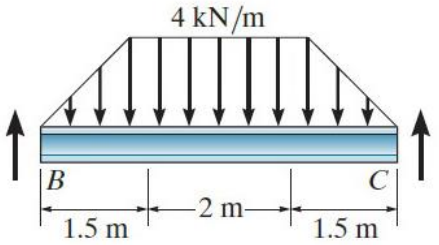
For example, the weight of a prismatic structural element (beam) is a uniformly distributed load.

Trapezoidal distributed load

A distributed line load with a linearly changing magnitude is referred to as trapezoidal distributed load.



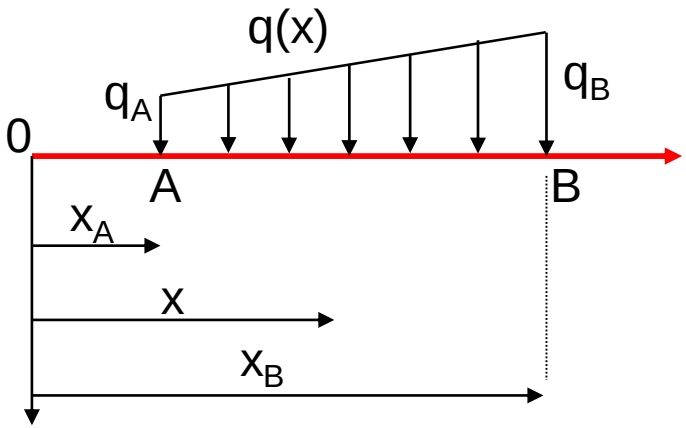




The load transferred from a slab or flat roof to a beam is trapezoidal

Trapezoidal distributed load

A distributed line load with a linearly changing magnitude is referred to as trapezoidal distributed load.



Function of the given load: **q(x) = ax+b**

where a and b are constant to be determined.

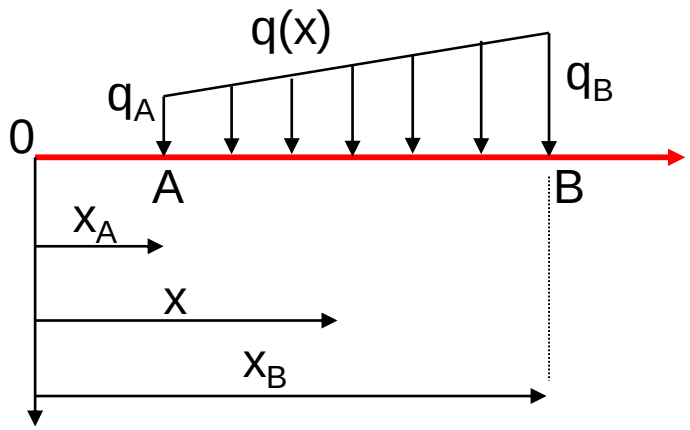
- For $x = x_A$, $q(x) = q_A$; for $x = x_B$, $q(x) = q_B$

$q_A = ax_A + b$ and $q_B = ax_B + b$

$$\mathbf{a} = \frac{\mathbf{q_B} - \mathbf{q_A}}{\mathbf{x_B} - \mathbf{x_A}} \qquad \mathbf{b} = \frac{\mathbf{q_A x_B} - \mathbf{q_B x_A}}{\mathbf{x_B} - \mathbf{x_A}}$$

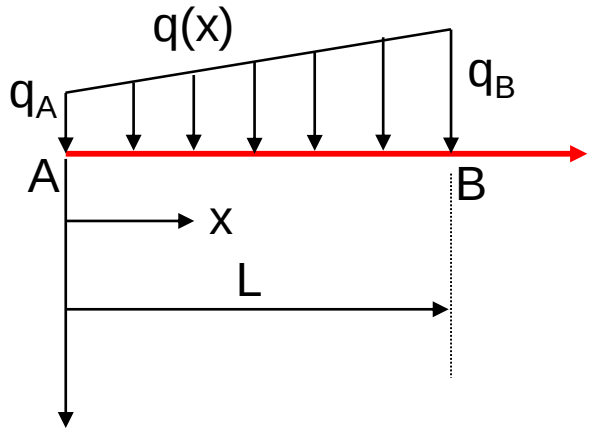
$$\mathbf{q(x)} = \frac{\mathbf{q_B} - \mathbf{q_A}}{\mathbf{x_B} - \mathbf{x_A}}\mathbf{x} + \frac{\mathbf{q_A x_B} - \mathbf{q_B x_A}}{\mathbf{x_B} - \mathbf{x_A}}$$

Trapezoidal distributed load



$$\mathbf{q(x) = \frac{q_B - q_A}{x_B - x_A} x + \frac{q_A x_B - q_B x_A}{x_B - x_A}}$$

- If the origin is located at the point A, for a trapezoidal distributed load of length L.



- $x_A = 0$, $x_B = L$

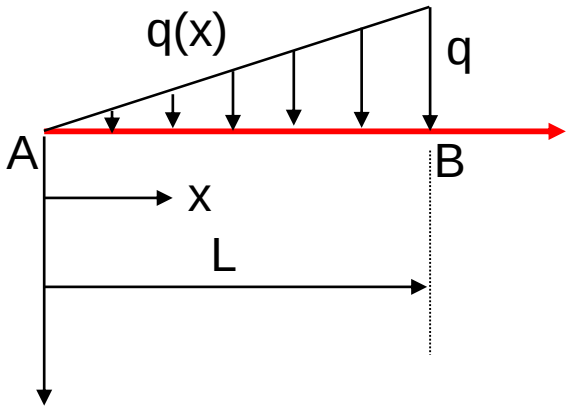
$$\mathbf{q(x) = \frac{q_B - q_A}{L} x + q_A}$$

Triangular distributed load

A distributed line load with a linearly changing magnitude starting from 0 or ending at 0 is referred to as triangular distributed load.

- *The load transferred from slab to a beam*
- *Hydrostatic pressure*
- *Soil effects*

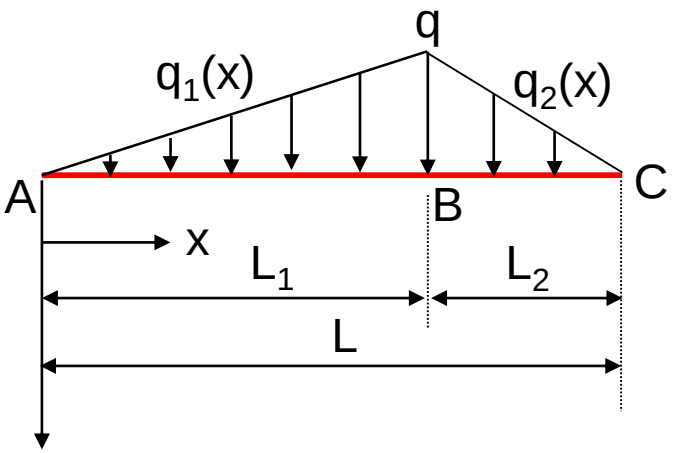
are examples of triangular distributed load.



- Considering trapezoidal distributed load function:
  - $q_A = 0$ and $q_B = q$

$$\mathbf{q(x) = \frac{q}{L} x}$$

Triangular distributed load



- For $0 \le x \le L_1$

$$\mathbf{q_1(x) = \frac{q}{L_1} x}$$

- For $L_1 < x \le L$

The functional form can be written by introducing the boundary conditions

$x = x_B \Rightarrow q_2(x) = q$ and $x = x_C \Rightarrow q(x) = 0$

$q = ax_B + b$ and $0 = ax_C + b$

$$\mathbf{a = -\frac{q}{x_C - x_B}} \qquad \mathbf{b = q\frac{x_C}{x_C - x_B}}$$

$x_C = L$ and $x_C - x_B = L_2 \Rightarrow$ $\mathbf{a = -\frac{q}{L_2}}$ $\mathbf{b = q\frac{L}{L_2}}$

$$\mathbf{q_2(x) = -\frac{q}{L_2} x + q\frac{L}{L_2}} \Rightarrow \mathbf{q_2(x) = q\frac{L - x}{L_2}}$$

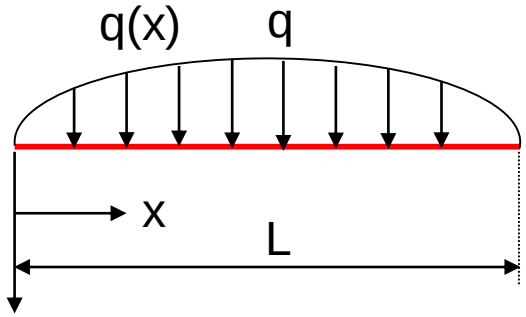
Parabolic distributed load

Three forms of parabolic distributed load are given below.

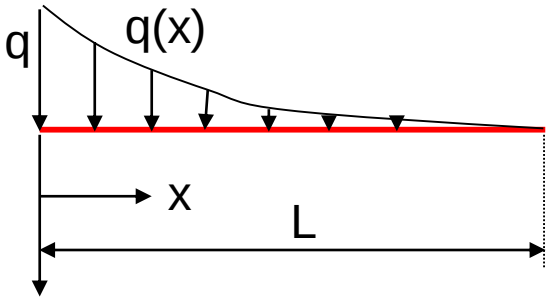
General functional form of a parabol is;

$q(x) = ax^2 + bx + c$

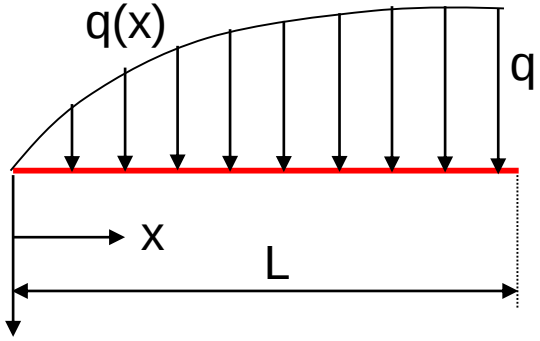
a, b and c are constants and these constants are estimated with introducing the boundary conditions into the general functional form



$$\mathbf{q(x) = \frac{4q}{L^2}(L - x)x}$$

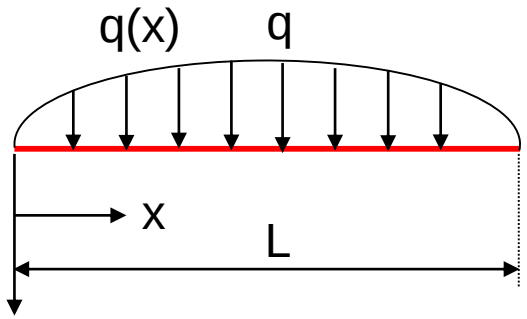


$$\mathbf{q(x) = \frac{q}{L^2}(L - x)^2}$$



$$\mathbf{q(x) = \frac{q}{L^2}(2L - x)x}$$

Parabolic distributed load



General functional form of a parabol is;

$q(x) = ax^2 + bx + c$

Boundary conditions:

x = 0 ; q(x) = 0 x = L ; q(x) = 0 x = L/2 ; q(x) = q

$$0 = a.0 + b.0 + c$$

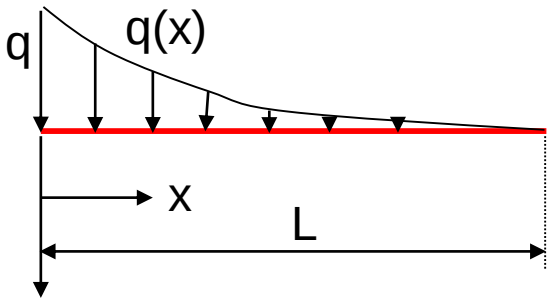
$$0 = aL^2 + bL + c$$

$$q = a\left(\frac{L}{2}\right)^2 + b\left(\frac{L}{2}\right) + c$$

$$\mathbf{a = -\frac{4q}{L^2}} \qquad \mathbf{b = -\frac{4q}{L}} \qquad \mathbf{c = 0}$$

$$\mathbf{q(x) = \frac{4q}{L^2}(L - x)x}$$

Parabolic distributed load



General functional form of a parabol is;

$q(x) = ax^2 + bx + c$

Boundary conditions:

$x = 0 ; q(x) = q \quad x = L ; q(x) = 0 \quad x = L ; dq(x)/dx = 0$

$q = a.0 + b.0 + c$

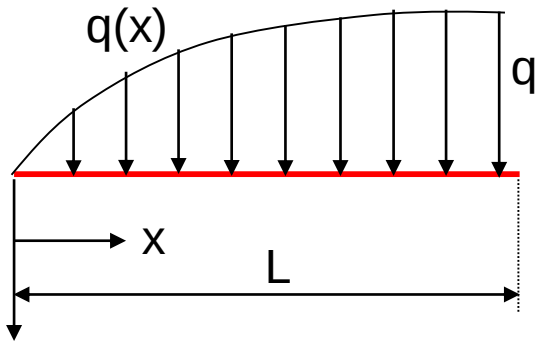
$0 = aL^2 + bL + c$

$0 = 2aL + b$

$$\mathbf{a = \frac{q}{L^2}} \qquad \mathbf{b = -\frac{2q}{L}} \qquad \mathbf{c = q}$$

$$\mathbf{q(x) = \frac{q}{L^2}(L - x)^2}$$

Parabolic distributed load



General functional form of a parabol is;

$q(x) = ax^2 + bx + c$

Boundary conditions:

$x = 0 ; q(x) = 0 \quad x = L ; q(x) = q \quad x = L ; dq(x)/dx = 0$

$0 = a.0 + b.0 + c$

$q = aL^2 + bL + c$

$0 = 2aL + b$

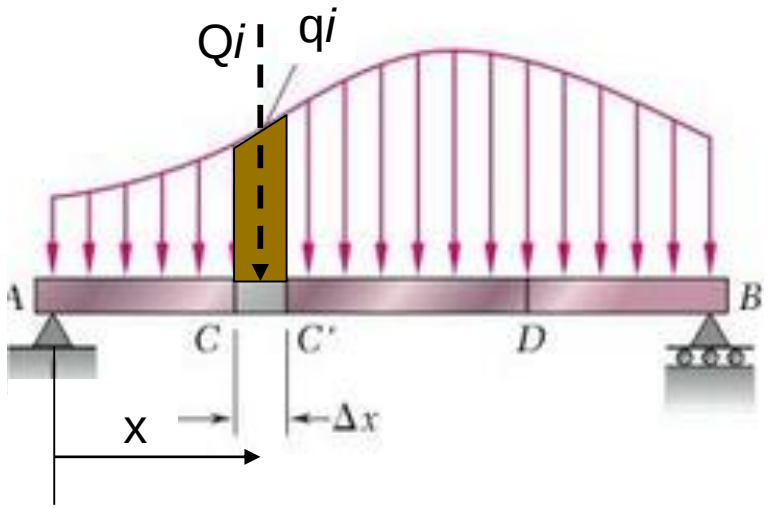
$$\mathbf{a = -\frac{q}{L^2}} \qquad \mathbf{b = \frac{2q}{L}} \qquad \mathbf{c = 0}$$

$$\mathbf{q(x) = \frac{q}{L^2}(2L - x)x}$$

Irregularly distributed load

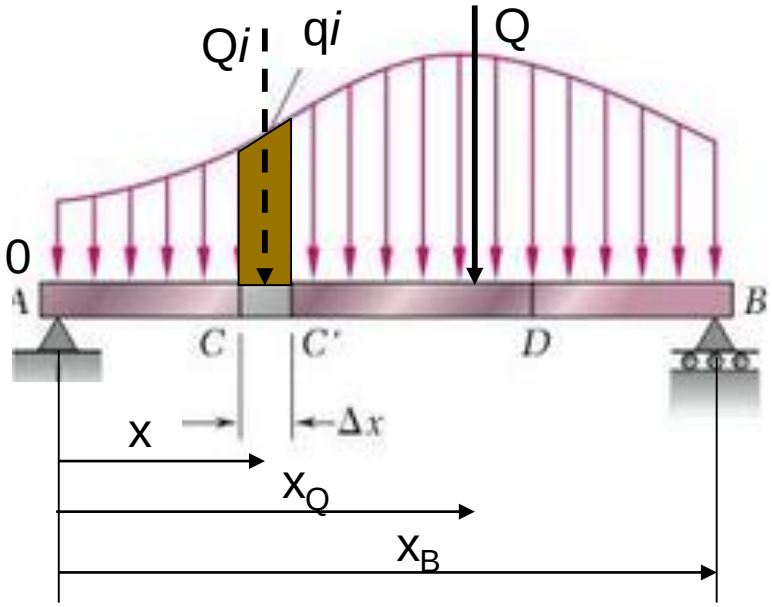
A distributed load whose functional form cannot be written is referred to as a irregularly distributed load.

This type of loads can be handled by dividing the load into infinitesimal pieces of length Δx and calculating the resultant point load acting on each piece individually.



$$Q_i = q_i . \Delta x$$

Resultant of distributed load



The magnitude of the distributed load at x is:

$$q(x) = \lim_{\Delta x \to 0} \frac{\Delta Q}{\Delta x}$$

ΔQ is the resultant load acting over the length Δx.

In other words, the magnitude of a distributed load is the value of load over unit length or unit area.

The area of the distributed line load diagram (or the volume of the distributed area load diagram) is the resultant of distributed load.

Resultant of distributed load

The resultant load

For irregularly distributed loads:

$$Q = \sum_{i=1}^{n} q(x_i)\Delta x$$

For regularly distributed loads:

$$Q = \int_{x_A}^{x_B} q(x)dx$$

Location of the Resultant of Distributed Load

The equilibrium condition requires that the moment due to the distributed load with respect to any point should be to the moment due to the resultant load with respect to the same point.

Let's write the moment with respect to 0:

- For irregularly distributed loads,

$$x_Q \sum \Delta Q_i = \sum (\Delta Q_i x) \rightarrow x_Q = \frac{\sum (\Delta Q_i x)}{\sum \Delta Q_i}$$

x_Q: the length of point that resultant load effects (center of gravity of distributed load)

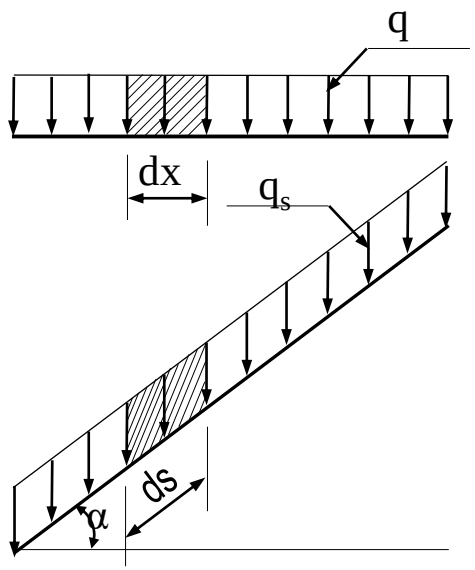
- For regularly distributed loads,

$$x_Q \int_{x_A}^{X_B} q(x)dx = \int_{x_A}^{x_B} q(x)xdx \rightarrow x_Q = \frac{\int_{x_A}^{x_B} q(x)xdx}{\int_{x_A}^{x_B} q(x)dx}$$

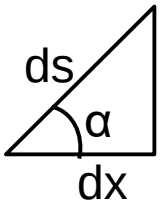
Transformation of Distributed Loads

Vertical Loads:

Let's transform the distributed load acting on a horizontal plane into a distributed load acting on an inclined plane in such a way that the magnitudes and locations of the resultant loads are the same.



$$q.dx = q_s.ds \qquad q_s = q\frac{dx}{ds}$$



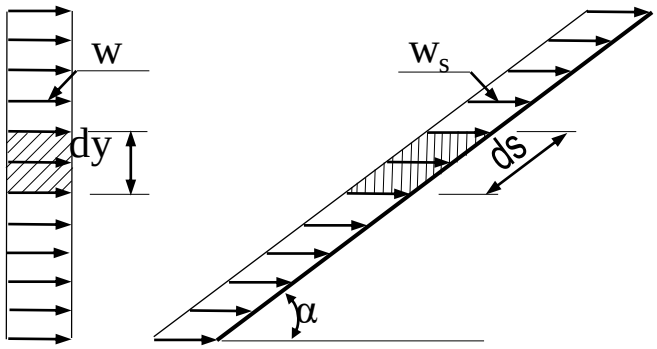
$$cos \propto = \frac{dx}{ds}$$

$$q_s = qcos \propto \qquad q = \frac{q_s}{cos \propto}$$

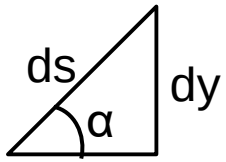
Transformation of Distributed Loads

Horizontal Loads:

Let's transform the distributed load acting on a vertical plane into a distributed load acting on an inclined plane in such a way that the magnitudes and locations of the resultant loads are the same.



$$w.dy = w_s.ds \qquad w_s = w\frac{dy}{ds}$$

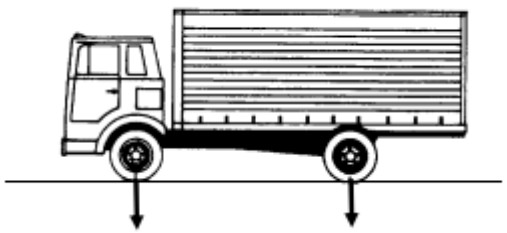


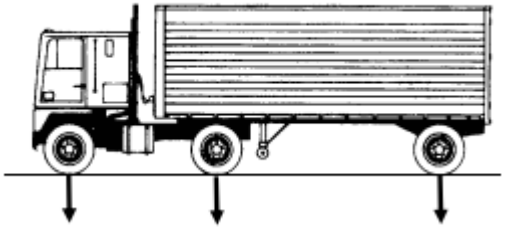
$$sin \propto = \frac{dy}{ds}$$

$$w_s = w sin \propto \qquad w = \frac{w_s}{sin \propto}$$

Moving Loads

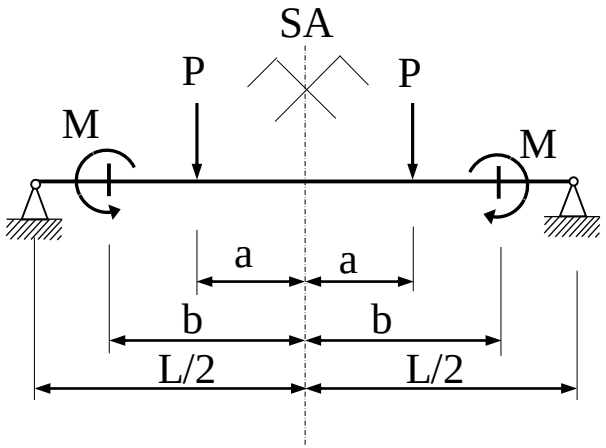
The loads representing vehicles on highway or railway bridges and crane loads are moving loads.





Symmetrical Loadings

Consider a geometrically and mechanically symmetrical structure. If the loads, located at equal distances to the symmetry axis on both sides, are equal and act in the same direction and the moments located at equal distances to the symmetry axis on both sides are equal and act in the opposite directions, then these loads are referred to as symmetrical loads.



Antimetrical Loadings

Consider a geometrically and mechanically symmetrical structure. If the loads, located at equal distances to the symmetry axis on both sides, are equal and act in the opposite directions and the moments located at equal distances to the symmetry axis on both sides are equal and act in the same direction, then these loads are referred to as antimetrical loads.

